

第一章 集合习题答案

习题 1.1 (Exercise 1.1)

1. 答案: (1) $\in, \notin, \notin, \notin, \notin$; (2) $\in, \notin, \in, \in, \in, \notin, \in, \notin$; (3) $\in, \in, \in, \in, \in, \in, \notin, \in, \in, \in, \in, \in$; (4) $\in, \in, \in, \in, \in, \in, \in, \in, \in$

解析: (1) $\mathbb{N}$ 是自然数集 (正整数和 0), 5 是自然数, 其余均不是; (2) $\mathbb{Z}$ 是整数集, $\sqrt{4} = 2$ 、 $(\sqrt{5})^2 = 5$ 是整数, 其余非整数; (3) $\mathbb{Q}$ 是有理数集 (分数和整数), $\sqrt{3}$ 、 -3π 是无理数, 其余均为有理数; (4) $\mathbb{R}$ 是实数集, 所有数均为实数。

Answer: (1) $\in, \notin, \notin, \notin, \notin$; (2) $\in, \notin, \in, \in, \in, \notin, \in, \notin$; (3) $\in, \in, \in, \in, \in, \in, \notin, \in, \in, \in, \in, \in$; (4) $\in, \in, \in, \in, \in, \in, \in, \in, \in$

Analysis: (1) $\mathbb{N}$ is the set of natural numbers (positive integers and 0), 5 is a natural number, and the rest are not; (2) $\mathbb{Z}$ is the set of integers, $\sqrt{4} = 2$ and $(\sqrt{5})^2 = 5$ are integers, and the rest are non-integers; (3) $\mathbb{Q}$ is the set of rational numbers (fractions and integers), $\sqrt{3}$ and -3π are irrational numbers, and the rest are rational numbers; (4) $\mathbb{R}$ is the set of real numbers, and all numbers are real numbers.

2. 答案: C

解析: 集合 $A = \{x \mid -5 < x \leq 5\}$, $\frac{5}{3} \approx 1.67$ 、 $-\frac{7}{3} \approx -2.33$ 、4.3 均满足 $-5 < x \leq 5$, -5 不满足 (左端点不包含), 故 C 正确。

Answer: C

Analysis: Set $A = \{x \mid -5 < x \leq 5\}$, $\frac{5}{3} \approx 1.67$, $-\frac{7}{3} \approx -2.33$, and 4.3 all satisfy $-5 < x \leq 5$, while -5 does not (the left endpoint is not included), so C is correct.

3. 答案: B

解析：集合 $B = \{x \in \mathbb{Z} \mid -4 \leq x < 2\}$ ，元素为 $-4, -3, -2, -1, 0, 1$ ； $-4 \in B$ ， -2.5 不是整数故 $\notin B$ ， $2 \notin B$ ， 1.5 不是整数故 $\notin B$ ，故 B 正确。

Answer: B

Analysis: Set $B = \{x \in \mathbb{Z} \mid -4 \leq x < 2\}$, whose elements are $-4, -3, -2, -1, 0, 1$; $-4 \in B$, -2.5 is not an integer so $\notin B$, $2 \notin B$, 1.5 is not an integer so $\notin B$, so B is correct.

4. 答案：B

解析： $\mathbb{Z}^+$ 是正整数集，满足 $x \leq 4$ 的正整数为 $1, 2, 3, 4$ ，故集合为 $\{1, 2, 3, 4\}$ 。

Answer: B

Analysis: $\mathbb{Z}^+$ is the set of positive integers, and the positive integers satisfying $x \leq 4$ are $1, 2, 3, 4$, so the set is $\{1, 2, 3, 4\}$.

5. 答案：B

解析：A 选项，若 $a = 2, b = 3$ ，则 $a - b = -1 \notin \mathbb{N}$ ，错误；B 选项，正整数属于有理数（有理数包含整数），正确；C 选项， $0.5 \geq 0$ 但 $\notin \mathbb{N}$ ，错误；D 选项，整数属于有理数，错误。

Answer: B

Analysis: Option A: If $a = 2, b = 3$, then $a - b = -1 \notin \mathbb{N}$, which is wrong; Option B: Positive integers belong to rational numbers (rational numbers include integers), which is correct; Option C: $0.5 \geq 0$ but $\notin \mathbb{N}$, which is wrong; Option D: Integers belong to rational numbers, which is wrong.

习题 1.2 (Exercise 1.2)

1. 答案：(1) $\in$; (2) $\nsubseteq$; (3) $\nsubseteq$; (4) $\nsubseteq$; (5) $=$; (6) $\notin$; (7) $\nsubseteq$; (8) $\nsubseteq$; (9) $\nsubseteq, \nsubseteq$, $\nsubseteq$

解析：(1) 3 是集合的元素，用 $\in$ ；(2) $\{1,2\}$ 是子集且不等于原集合，用 $\subseteq$ ；(3) 前一集合范围完全在后者内且不相等，用 $\subsetneq$ ；(4) 空集是任何非空集合的真子集；(5) 方程 $x^2 + 1 = 0$ 无实根，解集为空集，故相等；(6) 3 不小于 3，故 $\notin$ ；(7) $\{1,3\}$ 中 3 不在 $\{1,2,4\}$ 内，故 $\not\subset$ ；(8) 前一集合有 4 不在后者内，故 $\not\subset$ ；(9) 正整数集 $\subset$ 自然数集 $\subset$ 整数集 $\subset$ 有理数集 $\subset$ 实数集。

Answer: (1) $\in$; (2) $\subseteq$; (3) $\subsetneq$; (4) $\subseteq$; (5) $=$; (6) $\notin$; (7) $\not\subset$; (8) $\not\subset$; (9) $\subset$, $\subset$, $\subset$, $\subset$

Analysis: (1) 3 is an element of the set, so use $\in$; (2) $\{1,2\}$ is a subset and not equal to the original set, so use $\subseteq$; (3) The range of the first set is completely within the latter and not equal, so use $\subsetneq$; (4) The empty set is a proper subset of any non-empty set; (5) The equation $x^2 + 1 = 0$ has no real roots, and its solution set is the empty set, so they are equal; (6) 3 is not less than 3, so $\notin$; (7) 3 in $\{1,3\}$ is not in $\{1,2,4\}$, so $\not\subset$; (8) 4 in the first set is not in the latter, so $\not\subset$; (9) Positive integer set $\subset$ natural number set $\subset$ integer set $\subset$ rational number set $\subset$ real number set.

2. 答案：C

解析：A 选项，2 是元素，应用 $\in$ ，不是 $\subseteq$ ；B 选项， $\{2\}$ 是集合，应用 $\subseteq$ ，不是 $\in$ ；C 选项，3 不小于 3，故 $\notin \{x \mid -3 \leq x < 3\}$ ，正确；D 选项，空集不含元素， $\{0\}$ 含 0，不相等，错误。

Answer: C

Analysis: Option A: 2 is an element, so $\in$ should be used instead of $\subseteq$; Option B: $\{2\}$ is a set, so $\subseteq$ should be used instead of $\in$; Option C: 3 is not less than 3, so $\notin \{x \mid -3 \leq x < 3\}$, which is correct; Option D: The empty set has no elements, while $\{0\}$ contains 0, so they are not equal, which is wrong.

3. 答案: C

解析: 集合间不能用“ $<$ ”比较 (A 错误); M 是集合, 不是 N 的元素 (B 错误); M 中元素 2 在 N 中, 故 $M \subseteq N$ (C 正确); N 中 0, 4 不在 M 中, 故 $N \not\subseteq M$ (D 错误)。

Answer: C

Analysis: Sets cannot be compared with “ $<$ ” (A is wrong); M is a set, not an element of N (B is wrong); The element 2 in M is in N , so $M \subseteq N$ (C is correct); 0 and 4 in N are not in M , so $N \not\subseteq M$ (D is wrong).

4. 答案: D

解析: A 选项, $\pi \approx 3.14159 \neq 3.14$, 集合不相等; B 选项, M 是数集, N 是点集, 类型不同; C 选项, $M = \{0, 1\}$ (自然数含 0), 与 $N = \{1\}$ 不相等; D 选项, $|- \sqrt{4}| = 2$, $N = \{3, 1, 2\}$, 与 M 相等。

Answer: D

Analysis: Option A: $\pi \approx 3.14159 \neq 3.14$, so the sets are not equal; Option B: M is a number set, and N is a point set, which are of different types; Option C: $M = \{0, 1\}$ (natural numbers include 0), which is not equal to $N = \{1\}$; Option D: $|- \sqrt{4}| = 2$, $N = \{3, 1, 2\}$, which is equal to M .

5. 答案: 所有子集:

$\emptyset, \{1\}, \{3\}, \{5\}, \{7\}, \{1, 3\}, \{1, 5\}, \{1, 7\}, \{3, 5\}, \{3, 7\}, \{5, 7\}, \{1, 3, 5\}, \{1, 3, 7\}, \{1, 5, 7\}, \{3, 5, 7\}, \{1, 3, 5, 7\}$
; 真子集: 除 $\{1, 3, 5, 7\}$ 外的所有子集

解析: 含 n 个元素的集合有 2^n 个子集, 本题 $n = 4$, 故 16 个子集; 真子集是除集合本身外的子集, 共 15 个。

Answer: All subsets:

$\emptyset, \{1\}, \{3\}, \{5\}, \{7\}, \{1, 3\}, \{1, 5\}, \{1, 7\}, \{3, 5\}, \{3, 7\}, \{5, 7\}, \{1, 3, 5\}, \{1, 3, 7\}, \{1, 5, 7\}, \{3, 5, 7\}, \{1, 3, 5, 7\}$

; Proper subsets: All subsets except $\{1,3,5,7\}$

Analysis: A set with n elements has 2^n subsets. In this problem, $n = 4$, so there are 16 subsets; Proper subsets are subsets except the set itself, totaling 15.

习题 1.3 (Exercise 1.3)

1. 答案: $A \cup B = \{-1, 1, 2, 3, 4, 5, 6\}$, $A \cap B = \{-1\}$

解析: 并集是两集合所有元素 (去重), 交集是两集合公共元素, 两集合公共元素仅 -1 。

Answer: $A \cup B = \{-1, 1, 2, 3, 4, 5, 6\}$, $A \cap B = \{-1\}$

Analysis: The union is all elements of the two sets (removing duplicates), and the intersection is the common elements of the two sets. The only common element of the two sets is -1 .

2. 答案: $A \cup B = \{x \mid -5 \leq x \leq 4\}$, $A \cap B = \{x \mid -3 < x < 2\}$

解析: 并集取范围 “左到最左, 右到最右” (-5 到 4), 交集取范围 “左到较右, 右到较左” (-3 到 2)。

Answer: $A \cup B = \{x \mid -5 \leq x \leq 4\}$, $A \cap B = \{x \mid -3 < x < 2\}$

Analysis: The union takes the range “from the leftmost to the rightmost” (-5 to 4), and the intersection takes the range “from the right of the two left endpoints to the left of the two right endpoints” (-3 to 2).

3. 答案: $A \cup B = \{x \mid x \geq 0\}$, $A \cap B = \{x \mid x > 4\}$

解析: B 是 A 的子集 ($x > 4$ 必满足 $x \geq 0$), 故并集为 A , 交集为 B 。

Answer: $A \cup B = \{x \mid x \geq 0\}$, $A \cap B = \{x \mid x > 4\}$

Analysis: B is a subset of A (if $x > 4$, it must satisfy $x \geq 0$), so the union is A and the intersection is B .

4. 答案: $A \cup B = \{x \mid x \leq 3\}$, $A \cap B = \{x \mid x < -1\}$

解析: A 是 B 的子集 ($x < -1$ 必满足 $x \leq 3$), 故并集为 B , 交集为 A 。

Answer: $A \cup B = \{x \mid x \leq 3\}$, $A \cap B = \{x \mid x < -1\}$

Analysis: A is a subset of B (if $x < -1$, it must satisfy $x \leq 3$), so the union is B and the intersection is A .

5. 答案: $A \cup B = \{2\}$, $A \cap B = \emptyset$

解析: A 中方程 $x^2 + 4 = 0$ 无实根, 故 $A = \emptyset$; B 中方程 $2x - 4 = 0$ 解为 $x = 2$, 故 $B = \{2\}$;

空集与 $\{2\}$ 的并集为 $\{2\}$, 交集为空集。

Answer: $A \cup B = \{2\}$, $A \cap B = \emptyset$

Analysis: The equation $x^2 + 4 = 0$ in A has no real roots, so $A = \emptyset$; The solution of the equation $2x - 4 = 0$ in B is $x = 2$, so $B = \{2\}$; The union of the empty set and $\{2\}$ is $\{2\}$, and the intersection is the empty set.

6. 答案: $C_U A = \{0, 4, 10\}$, $C_U B = \{2, 8\}$, $(C_U A) \cap (C_U B) = \emptyset$

解析: 补集是全集中不属于该集合的元素: U 中除 A 的元素, 剩余 $0, 4, 10$; 除 B 的元素, 剩余 $2, 8$; 两补集无公共元素, 交集为空集。

Answer: $C_U A = \{0, 4, 10\}$, $C_U B = \{2, 8\}$, $(C_U A) \cap (C_U B) = \emptyset$

Analysis: The complement is the elements in the universal set that do not belong to the set: In U , except the elements of A , the remaining elements are $0, 4, 10$; Except the elements of B , the remaining elements are $2, 8$; The two complements have no common elements, so the intersection is the empty set.

7. 答案: $C_U A = \{x \mid -5 \leq x < -2 \text{ 或 } 2 \leq x < 5\}$

解析: 补集是 U 中不属于 A 的范围: U 为 $-5 \leq x < 5$, A 为 $-2 \leq x < 2$, 故补集为 $-5 \leq x < -2$ 或 $2 \leq x < 5$

-2 或 $2 \leq x < 5$ 。

Answer: $C_U A = \{x \mid -5 \leq x < -2 \text{ or } 2 \leq x < 5\}$

Analysis: The complement is the range in U that does not belong to A : U is $-5 \leq x < 5$, A is $-2 \leq x < 2$, so the complement is $-5 \leq x < -2$ or $2 \leq x < 5$.

8. 答案: $C_U A = \{-5, -4, -3, 2, 3, 4, 5\}$

解析: $U = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$, $A = \{-2, -1, 0, 1\}$, 补集为 U 中除 A 外的元素。

Answer: $C_U A = \{-5, -4, -3, 2, 3, 4, 5\}$

Analysis: $U = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$, $A = \{-2, -1, 0, 1\}$, and the complement is the elements in U except A .

9. 答案: $A \cup B = (-4, 5]$, $A \cap B = [-1, 2)$, $C_U A = (-\infty, -4] \cup [2, +\infty)$, $C_U B = (-\infty, -1) \cup (5, +\infty)$

解析: 并集: $A = (-4, 2)$ 、 $B = [-1, 5]$, 合并范围为 $(-4, 5]$; 交集: 公共范围为 $[-1, 2)$;

补集: $C_U A$ 是 $\mathbb{R}$ 中除 $(-4, 2)$ 的部分, $C_U B$ 是 $\mathbb{R}$ 中除 $[-1, 5]$ 的部分。

Answer: $A \cup B = (-4, 5]$, $A \cap B = [-1, 2)$, $C_U A = (-\infty, -4] \cup [2, +\infty)$, $C_U B = (-\infty, -1) \cup (5, +\infty)$

Analysis: Union: $A = (-4, 2)$ and $B = [-1, 5]$, the combined range is $(-4, 5]$;

Intersection: The common range is $[-1, 2)$; Complement: $C_U A$ is the part of $\mathbb{R}$ except $(-4, 2)$, and $C_U B$ is the part of $\mathbb{R}$ except $[-1, 5]$.

10. 答案: A

解析: 先求 $M \cup N = \{0, 1, 2, 3, 4, 5, 7\}$, 再求 U 中不属于该并集的元素: $6, 8$, 故 $C_U(M \cup N) = \{6, 8\}$ 。

Answer: A

Analysis: First find $M \cup N = \{0, 1, 2, 3, 4, 5, 7\}$, then find the elements in U that do not

belong to this union: 6,8, so $C_U(M \cup N) = \{6,8\}$.

11. 答案: C

解析: 先求 $C_U A = \{1,6,7\}$ (U 中除 A 的元素), 再求 B 与该补集的交集: $B = \{2,3,6,7\}$, 公共元素为 6,7, 故 $B \cap (C_U A) = \{6,7\}$ 。

Answer: C

Analysis: First find $C_U A = \{1,6,7\}$ (elements in U except A), then find the intersection of B and this complement: $B = \{2,3,6,7\}$, and the common elements are 6,7, so $B \cap (C_U A) = \{6,7\}$.

自测题 (Self-test)

1. 答案: (1) $\notin$; (2) $\subsetneq$; (3) $\subsetneq$; (4) $\subsetneq$; (5) $\notin$; (6) $\in$; (7) $\in, \notin, \in$; (8) $\nsubseteq$

解析: (1) 5 不在 $\{-4,0,4\}$ 中; (2) $\{0,3\}$ 是子集且不相等; (3) 前一范围在后者内且不相等; (4) 空集是 $\{x \mid 3x + 6 = 0\} = \{-2\}$ 的真子集; (5) 2 不小于 2, 故 $\notin$; (6) $0 \leq 5$, 故 $\in$; (7) $\sqrt{25} = 5 \in \mathbb{Q}$, $\frac{4}{7}$ 是分数 $\notin \mathbb{Z}$, $-5.2 \in \mathbb{R}$; (8) $\{-1,0,1\}$ 中 0,1 不在 $\{-1,2,3\}$ 内, 故 $\nsubseteq$ 。

Answer: (1) $\notin$; (2) $\subsetneq$; (3) $\subsetneq$; (4) $\subsetneq$; (5) $\notin$; (6) $\in$; (7) $\in, \notin, \in$; (8) $\nsubseteq$

Analysis: (1) 5 is not in $\{-4,0,4\}$; (2) $\{0,3\}$ is a subset and not equal; (3) The first range is within the latter and not equal; (4) The empty set is a proper subset of $\{x \mid 3x + 6 = 0\} = \{-2\}$; (5) 2 is not less than 2, so $\notin$; (6) $0 \leq 5$, so $\in$; (7) $\sqrt{25} = 5 \in \mathbb{Q}$, $\frac{4}{7}$ is a fraction so $\notin \mathbb{Z}$, $-5.2 \in \mathbb{R}$; (8) 0 and 1 in $\{-1,0,1\}$ are not in $\{-1,2,3\}$, so $\nsubseteq$.

2. 答案: (1) $A \cup B = \{-1,0,2,3,4,6\}$, $A \cap B = \{0\}$; (2) $A \cup B = \{x \mid -4 \leq x \leq 2\}$, $A \cap B = \{x \mid -2 < x < 1\}$; (3) $C_U A = \{-1,3,7\}$, $C_U B = \{-1,0,3\}$, $(C_U A) \cap (C_U B) = \{-1,3\}$; (4) $C_U A = \{4,5,6,7\}$; (5) $A \cup B = (1,7]$, $A \cap B = [3,5]$, $C_U A = (0,3) \cup (5, +\infty)$, $C_U B =$

$(7, +\infty)$; (6) $(C_U A) \cap B = \{x \mid 2 < x < 5\}$; (7) $A \cap B = \{-1, 0\}$; (8) $C_U(A \cup B) = \{x \mid 0 < x < 1\}$; (9) 4

解析: (1) 并集取两集合所有元素, 交集取公共元素 0; (2) 并集范围-4到2, 交集范围-2到1; (3) $C_U A$ 是 U 除 A 的元素, $C_U B$ 是 U 除 B 的元素, 交集取公共元素-1,3; (4) $U = \{0, 1, 2, 3, 4, 5, 6, 7\}$, $A = \{0, 1, 2, 3\}$, 补集为 $\{4, 5, 6, 7\}$; (5) 并集合并 $(1, 7]$, 交集取 $[3, 5]$, 补集按范围计算; (6) $C_U A = (-\infty, 0) \cup (2, +\infty)$, 与 $B = [1, 5)$ 的交集为 $(2, 5)$; (7) $B = \{1, 0, -1\}$ (代入 $x \in A$ 计算 y), 与 A 的交集为 $\{-1, 0\}$; (8) $A \cup B = (-\infty, 0] \cup [1, +\infty)$, 补集为 $(0, 1)$; (9) $A \cap B = \{2, 3\}$ (含 2 个元素), 子集个数为 $2^2 = 4$ 。

Answer: (1) $A \cup B = \{-1, 0, 2, 3, 4, 6\}$, $A \cap B = \{0\}$; (2) $A \cup B = \{x \mid -4 \leq x \leq 2\}$, $A \cap B = \{x \mid -2 < x < 1\}$; (3) $C_U A = \{-1, 3, 7\}$, $C_U B = \{-1, 0, 3\}$, $(C_U A) \cap (C_U B) = \{-1, 3\}$; (4) $C_U A = \{4, 5, 6, 7\}$; (5) $A \cup B = (1, 7]$, $A \cap B = [3, 5]$, $C_U A = (0, 3) \cup (5, +\infty)$, $C_U B = (7, +\infty)$; (6) $(C_U A) \cap B = \{x \mid 2 < x < 5\}$; (7) $A \cap B = \{-1, 0\}$; (8) $C_U(A \cup B) = \{x \mid 0 < x < 1\}$; (9) 4

Analysis: (1) The union takes all elements of the two sets, and the intersection takes the common element 0; (2) The union range is from -4 to 2, and the intersection range is from -2 to 1; (3) $C_U A$ is the elements of U except A , $C_U B$ is the elements of U except B , and the intersection takes the common elements -1, 3; (4) $U = \{0, 1, 2, 3, 4, 5, 6, 7\}$, $A = \{0, 1, 2, 3\}$, so the complement is $\{4, 5, 6, 7\}$; (5) The union combines to $(1, 7]$, the intersection takes $[3, 5]$, and the complement is calculated by range; (6) $C_U A = (-\infty, 0) \cup (2, +\infty)$, and its intersection with $B = [1, 5)$ is $(2, 5)$; (7) $B = \{1, 0, -1\}$ (calculate y by substituting $x \in A$), and its intersection with A is $\{-1, 0\}$; (8) $A \cup B = (-\infty, 0] \cup [1, +\infty)$, so the complement is $(0, 1)$; (9) $A \cap B = \{2, 3\}$ (containing 2

elements), and the number of subsets is $2^2 = 4$.